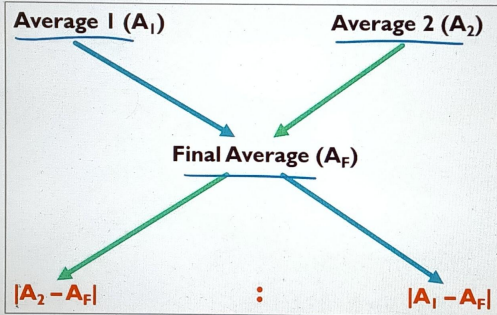




1. Exact soln to mixture problems

(Alligation is the shortcut)

CONCEPT - ALLIGATION



13. In what ratio must a grocer mix two varieties of pulses Rs.15 and Rs.20 per kg respectively so as to get a mixture worth Rs.16.50 per kg?

A) 3 : 7

B) 5 : 7

C) 7 : 3

D) 7 : 5

$$\begin{array}{l} A \quad B \\ 15 \quad 20 \\ x \quad y \\ 15x + 20y = 16.5 \\ x + y = 1 \\ 15x + 20y = 16.5x + 16.5y \\ 16.5x - 15x = 20y - 16.5y \\ 1.5x = 3.5y \\ \frac{x}{y} = \frac{3.5}{1.5} = \frac{35}{15} = \frac{7}{3} \end{array}$$

$$\begin{array}{r} 15 \quad 20 \\ \quad 16.5 \\ 3.5 \quad 1.5 \\ 35 \quad 15 \\ \hline 7 \quad 3 \end{array}$$

15. A dealer buys 11 kg of wheat at Rs.374 and mixes it with quality of wheat in the ratio of 4:3. The price of the resulting mixture is Rs.40 per kg. The price of the other quality of wheat is _____?

A) Rs.48/kg

B) Rs.50/kg

C) Rs.42/kg

D) Rs.32/kg

$$A_1 = \frac{374}{11} = 34$$

$$\begin{array}{r} 34 \quad ? \\ \quad 40 \\ \times 2 \quad 4 \quad : \quad 3 \quad \times 2 \\ \quad 8 \quad : \quad 6 \end{array}$$

$40 + 8 = 48$

$$\begin{array}{r} 46 \quad ? \\ \quad 40 \\ \times 2 \quad 4 \quad : \quad 3 \quad \times 2 \\ \quad 8 \quad : \quad 6 \end{array}$$

$40 - 8 = 32$

48 liya cuz $\frac{34 + 48}{2} = 40$, Similarly, idhar 32 liya $40 + 8 = 48$
32 nahi le sakte avg nahi chuz avg
40 x aayega. 40 aana chahiye

16. A shopkeeper has 50kg of rice. He sells a part of it at 20% profit and the rest at 40% profit. He gains 25% on the whole. Find the ratio of the two parts.

A) 3 : 1

B) 2 : 1

C) 1 : 3

D) 4 : 1

$$\begin{array}{r} 20\% \quad 40\% \\ \quad 25\% \\ 15 \quad : \quad 5 \\ 3 \quad : \quad 1 \end{array}$$

$$\begin{array}{r} 80\% \quad 60\% \\ \quad 75\% \\ 15 \quad : \quad 5 \\ 3 \quad : \quad 1 \end{array}$$

$$\begin{array}{r} 20\% P \quad 40\% L \quad 25\% L \\ 120\% \quad 60\% \\ \quad 75\% \\ 15 \quad : \quad 45 \\ 1 \quad : \quad 3 \end{array}$$

